



AlgoProfessor

AI R&D Solutions

Innovating the Future with Artificial Intelligence

3-Month Immersive Online Internship Programme 2026

100% Online • Live Mentorship • Global Cohort

From AI Foundations → Agentic AI → Industry Placement

90 ★ **18** ★ **22+** ★ **60+**
Days Milestones Skill Areas AI Tools

Four-Stage Learning Architecture

Phase 1 (Days 1–30): Foundations, ML, DL, NLP, Databases, RAG

Phase 2 (Days 31–60): LLM Engineering, Fine-tuning, Multimodal

Phase 3 (Days 61–90): Agentic AI, MCP, Multi-Agent, Deployment

Bridge (90+): Placement, Interview Prep, Portfolio Showcase

Programme Fee

INR 50,000 • USD 1,000

Inclusive of all materials, mentorship, certification, and placement bridge

Open to: B.Tech M.Tech Ph.D Post-Doctoral

✉ **ceo@algo professor.com** ★ **www.algo professor.com**

Batch 2026 • 100% Online • Daily GitHub Commits • Placement Support

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Message from the CEO

Dr. S. Satyanarayana

M.Tech (CSE), Ph.D. (Graph ML), Ph.D. (Cloud), DEng (AI & DL), FARSC (USA)

Founder & Chief AI Scientist, AlgoProfessor AI R&D Solutions

Editor-in-Chief, IJCM • 21+ Years • 60+ Papers • 9 Books • 8 Patents

Dear Future AI Professional,

Welcome to the **AlgoProfessor AI R&D Solutions 3-Month Online Internship 2026**, one of the most intensive hands-on AI training experiences available in India today, delivered **completely online** so you can join from anywhere in the world.

AlgoProfessor was founded with a singular vision: to close the gap between academic AI education and industry demand. Universities teach theory beautifully. Industry needs professionals who can build, ship, and iterate on intelligent systems at speed. That is what this programme delivers.

Over 90 days, you will write production-grade code every single day from your own laptop. You will build AI agents that browse the web, write and test software, conduct deep research, and manage autonomous workflows. You will work with the same tools that power the world's most advanced AI applications today: **GPT-5, Claude Opus 4.7, Llama 4, Gemini 2.5, LangGraph 1.0, CrewAI, AutoGen, and Anthropic MCP**.

Programme highlights:

- ✓ **100% online delivery** with live mentorship sessions and recorded lectures
- ✓ **18 milestone projects** including new tracks on Voice AI and Computer-Use agents
- ✓ **22+ skill categories** across the modern AI stack with 60+ tools
- ✓ **Post-90 Placement Bridge** covering interviews, portfolio polish, and negotiation
- ✓ **Global cohort access** with students and mentors across India, USA, Europe, and SEA
- ✓ **Affordable fee:** INR 50,000 for India and USD 1,000 for international applicants

The **daily GitHub submission requirement** is your daily commitment to yourself. The discipline of committing working, documented code every day, even when the task is hard, is what separates capable engineers from outstanding ones.

To students from B.Tech, M.Tech, Ph.D, and Post-Doctoral backgrounds, I encourage you to bring your domain expertise. AI is most powerful when combined with deep knowledge of medicine, law, finance, materials science, or any other field. Your domain knowledge becomes a genuine competitive advantage.

The AI revolution is not happening to us. We are building it, one commit at a time.

With the highest expectations,

Dr. S. Satyanarayana

Founder & Chief AI Scientist, AlgoProfessor AI R&D Solutions

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1 Programme Fee & Enrolment

Internship Programme Fee 2026 Batch

Region	Programme Fee	Includes
India	INR 50,000	Full 90-day curriculum
International	USD 1,000	Full 90-day curriculum

1.1 What is Included in the Fee

- **✓ Full 90-day curriculum:** all lectures, exercises, and 18 milestone projects
- **✓ Daily mentor reviews:** structured feedback every 10 days, peer code review
- **✓ Cloud credits:** starter credits for OpenAI, Anthropic, AWS, GCP, HuggingFace
- **✓ Project showcase:** flagship projects deployed on Hugging Face Spaces or Vercel
- **✓ Certification:** Completion Certificate, Letter of Recommendation for top performers
- **✓ Placement Bridge:** 4–6 weeks of interview prep, mock interviews, salary coaching
- **✓ Industry network access:** hiring panel demo day, recruiter introductions
- **✓ LinkedIn endorsements:** verified endorsements from AlgoProfessor leadership

1.2 Payment Schedule

Phase	Milestone	India	International
Enrolment	On confirmation of seat	INR 20,000	USD 400
Phase 1 End	Day 30 (after Milestone 5)	INR 15,000	USD 300
Phase 2 End	Day 60 (after Milestone 9)	INR 15,000	USD 300
Total		INR 50,000	USD 1,000

1.3 Scholarships & Discounts

➡ Available Concessions

- **Merit scholarship:** up to 50 percent off for top 5 percent of applicants (entrance test)
- **Early-bird discount:** 15 percent off for enrolments 30 days before batch start
- **Group enrolment:** 10 percent off for groups of 3 or more from same institution
- **Women-in-tech:** 20 percent off for women applicants
- **Need-based aid:** financial assistance for deserving students (proof required)

1.4 Refund Policy

Withdrawal Period	Refund
Before programme start	90 percent refund (10 percent admin fee)
Within Days 1–7	50 percent refund
After Day 7	No refund (programme materials already issued)

2 Programme Overview & Roadmap

This 90-day fully online programme takes interns from Python and Git fundamentals to production-grade Agentic AI, and onward to industry placement. Every 10 days a milestone project synthesises the learning. Each 30-day phase ends with a major capstone. Interns commit code to GitHub daily and receive mentor reviews every 10 days. A structured **Post-90 Placement Bridge** prepares you for AI engineering and research roles in India and overseas.

Programme Architecture

Phase	Days	Theme	Milestones	Reviews
Phase 1	1–30	Foundations: Git, Python, ML, DL, NLP, Databases, RAG	5	3
Phase 2	31–60	LLM Engineering, Fine-tuning, Multi-modal, Evaluation	5	3
Phase 3	61–90	Agentic AI, MCP, Multi-Agent, Deployment	8	3
Bridge	91+	Placement Prep, Interview Coaching, Portfolio	2	Continuous
Total			20	9+

2.1 GitHub Daily Submission Protocol

Submission Rules

- **Repository:** `algoprofessor-rd-internship-2026-<your-name>`
- **Deadline:** Push code and documentation by **5:00 PM IST** daily
- **Notification:** Email GitHub link to `ceo@algoprofessor.com`
- **Layout:** every day-folder must contain `README.md`, `requirements.txt`, `outputs/`, `tests/`
- **Commit format:** `[Day-N] Short descriptive title`
- **Quality:** PEP 8, type hints, docstrings, 70 percent test coverage minimum

3 Expanded Skill Matrix 2026

The 2026 programme covers **22+ skill categories** grouped into four pillars. Each skill is paired with the day(s) where you build it and the typical job role it unlocks.

3.1 Pillar 1: Engineering Foundations

Skill	What you learn	Days	Job Role
Version Control & DevOps	Git, GitHub Actions, Docker, Kubernetes, Helm, CI/CD, pre-commit hooks	1, 28, 84–87	DevOps Engineer
Python Engineering	Python 3.12+, type hints, async/await, pytest, Pydantic v2, uv	2–5	Backend Engineer
Data Engineering	NumPy, Pandas 2.x, Polars, Parquet, Arrow, DuckDB	3, 11–12	Data Engineer
Visualisation	Matplotlib, Seaborn, Plotly, Altair, Dash, Streamlit	4	Analytics Engineer
Databases	PostgreSQL, MongoDB, Redis, pgvector, SQLAlchemy 2.0 async	11–12	Database Engineer

3.2 Pillar 2: Classical AI & Modelling

Skill	What you learn	Days	Job Role
Machine Learning	Scikit-learn, XGBoost 2.x, LightGBM, CatBoost, SHAP, Optuna	6–7	ML Engineer
Deep Learning	PyTorch 2.x, torch.compile, FSDP, mixed precision	8–9	DL Researcher
Computer Vision	ResNet, EfficientNetV2, ViT, Swin V2, DINOv3, SAM 2, YOLOv11	9	CV Engineer
NLP & Transformers	Hugging Face, tokenisers, embeddings, BERT fine-tuning	10	NLP Engineer
Vector Databases	Chroma, FAISS, Pinecone, Qdrant, Weaviate, hybrid search	13	Search Engineer
RAG Systems	Naive, Advanced, Modular, HyDE, Graph RAG, RAGAS	14–15	RAG Engineer

3.3 Pillar 3: Generative & Agentic AI

Skill	What you learn	Days	Job Role
LLM API Engineering	GPT-5, Claude Opus 4.7, Gemini 2.5, LiteLLM, prompt caching	16–19	LLM Engineer

Skill	What you learn	Days	Job Role
Prompt Engineering	Zero/few-shot, CoT, ToT, ReAct, DSPy, self-consistency	17, 33	Prompt Engineer
Open-Source LLMs	Llama 4, Mistral, Phi-4, DeepSeek R1, Qwen 3, vLLM, Ollama	22, 39	LLM Deployer
Fine-tuning	LoRA, QLoRA, DPO, ORPO, KTO, GRPO, Unsloth, Axolotl	36–38	Fine-tuning Specialist
Multimodal AI	GPT-4o, Claude Vision, Gemini Vision, LLaVA 1.6	23, 42	Multimodal Engineer
Voice & Audio AI	Whisper v3, ElevenLabs, Cartesia, real-time voice agents	23, 50	Voice AI Engineer
Agentic Frameworks	LangChain, LangGraph 1.0, CrewAI, AutoGen, smolagents	61–72	Agentic AI Engineer
MCP Servers	Anthropic Model Context Protocol, custom server design	62, 76	MCP Developer
Browser & Computer Agents	Playwright, Stagehand, Claude Computer Use	64, 73–75	Automation Engineer

3.4 Pillar 4: MLOps, Safety & Specialisations

Skill	What you learn	Days	Job Role
MLOps & Observability	MLflow, W&B, LangSmith, Arize Phoenix, OpenTelemetry	26, 46	MLOps Engineer
LLM Evaluation	RAGAS, LM Eval Harness, MT-Bench, AgentBench, GAIA	44, 78	Evaluation Engineer
AI Safety & Guardrails	NeMo Guardrails, Llama Guard 3, Presidio, Garak red teaming	27, 47	Safety Engineer
AI Governance	EU AI Act, India DPDP, ISO 42001, model cards	47, 89	Governance Lead
Cloud & Deployment	AWS Bedrock, GCP Vertex AI, Azure AI Foundry, Modal	49, 84–87	Cloud Architect
Edge AI & Quantisation	GGUF, AWQ, GPTQ, ONNX, on-device Llama	39	Edge AI Engineer
Federated AI	Flower, Opacus, differential privacy	Elective	Privacy AI Engineer

4 Phase 1: Foundations (Days 1–30)

Phase 1 Focus Areas

Git & GitHub • Python 3.12+ • NumPy, Pandas, Polars • ML & DL • NLP • Databases • Vector Stores • RAG • LLM APIs • Prompt Engineering

4.1 Week 1: Version Control & Python (Days 1–5)

✓ Day 1: Git & GitHub Mastery

Learn: version control, branching, rebasing, PRs, code review, GitHub Actions, CLI, GPG signing.

Build: `day01_git_mastery/`, `branching_demo/`, `git_cheatsheet.md`.

✓ Day 2: Python 3.12+ & Modern Tooling

Learn: type hints, pattern matching, decorators, Pydantic v2. Tooling: `uv`, `ruff`, `mypy`, `pytest`.

Build: `core_python.py`, `data_structures_bench.py`, full test suite.

✓ Day 3: NumPy, Pandas 2.x, Polars & EDA

Learn: ndarray ops, Pandas 2.x with PyArrow, Polars for high-performance dataframes.

Build: `numpy_ops.py`, `eda_pipeline.py`, `polars_benchmark.py`.

✓ Day 4: Visualisation: Matplotlib, Seaborn, Plotly

Learn: publication-quality charts, interactive dashboards, accessibility-aware palettes.

Build: `static_charts.py`, `interactive_dashboard.py`.

✓ Day 5: OOP, File I/O & Project Environments

Learn: OOP patterns, `uv` dependency management, Docker basics, structured logging.

Build: `oop_design_patterns.py`, `file_manager.py`, `project_template/`.

4.2 Week 2: Machine Learning & Deep Learning (Days 6–10)

✓ Day 6: Scikit-learn & ML Pipelines

Learn: regression, classification, SHAP, Optuna 4.x for hyperparameter tuning.

Build: `ml_pipeline.py`, `model_comparison.ipynb`, `shap_analysis.py`.

✓ Day 7: Ensemble Methods & Feature Engineering

Learn: XGBoost 2.x, LightGBM, CatBoost, PCA, UMAP, stacking ensembles.

Build: `ensemble_models.py`, `feature_engineering.py`.

✓ Day 8: Neural Networks & PyTorch 2.x

Learn: NN from scratch, PyTorch 2.x with `torch.compile`, mixed precision, TensorBoard.

Build: `nn_from_scratch.py`, `pytorch_cnn.py` targeting 85% on CIFAR-10.

✓ Day 9: Transfer Learning & Modern Vision

Learn: fine-tune ResNet50, EfficientNetV2, ViT, Swin V2, DINOv3, SAM 2.

Build: `transfer_learning.py`, ONNX export, runtime benchmarks.

✓ Day 10: NLP & Hugging Face Transformers

Learn: tokenisation, embeddings, fine-tune DistilBERT/roBERTa, push to HF Hub.

Build: `text_preprocessing.py`, `bert_sentiment.py`, `hf_finetune.py`.

★ Milestone 1 (Day 10): Web Intelligence Synthesiser

AI-powered brochure generator. Scrapes company websites, classifies sections via NLP, synthesises a 2-page branded PDF via LLM. Streamlit UI with live progress, SQLite caching.

Stack: BeautifulSoup4, Playwright, OpenAI API, ReportLab, Streamlit, SQLite.

4.3 Week 3: Databases, Vector Stores & RAG (Days 11–15)

✓ Day 11: PostgreSQL, pgvector & SQLAlchemy 2.0

Learn: JOINS, CTEs, window functions, JSONB, pgvector, SQLAlchemy 2.0 async.

Build: `db_schema.sql`, `orm_models.py`, `async_queries.py`.

✓ Day 12: MongoDB, Redis & Caching

Learn: MongoDB aggregation, Atlas Search, Redis caching patterns, Celery queues.

Build: `mongo_crud.py`, `agg_pipeline.py`, `redis_cache.py`.

✓ Day 13: Vector Databases & Semantic Search

Learn: embeddings (OpenAI, Cohere v4, Voyage 3), HNSW vs IVF, hybrid search.

Build: `embedding_gen.py`, `vector_db_setup.py`, `hybrid_search.py`.

✓ Day 14: RAG Foundations

Learn: Naive vs Advanced vs Modular RAG, chunking strategies, RAGAS evaluation.

Build: `doc_processor.py`, `rag_pipeline.py`, `ragas_eval.py`.

✓ Day 15: Advanced RAG: HyDE, Graph RAG, Agentic RAG

Learn: HyDE, multi-query, Graph RAG with Neo4j, agentic RAG, streaming responses.

Build: `advanced_rag.py`, `graph_rag.py`, `streaming_qa_app.py`.

★ Milestone 2 (Day 15): Enterprise Knowledge Navigator

AI knowledge worker using advanced RAG over 50+ company documents. Graph RAG with Neo4j, multi-query and HyDE retrieval, Cohere reranking, 10-turn memory, live RAGAS dashboard.

Stack: LangChain, Pinecone, Neo4j, FastAPI, Streamlit, RAGAS.

4.4 Days 16–30: LLM APIs, Prompting & Phase 1 Capstones

Day	Topic	Key Tasks	Tools
16	OpenAI, Anthropic, Gemini APIs	SDK integration, cost tracking, prompt caching	OpenAI, Anthropic, Gemini
17	Prompt Engineering	Zero/few-shot, CoT, ToT, ReAct, DSPy	DSPy, PromptFoo
18	Function Calling & Structured Outputs	JSON schema, tool orchestration, Pydantic AI	Pydantic AI, Instructor
19	Conversational AI & Memory	Multi-turn chat, summary memory, Mem0	LangChain, Mem0, Redis
20	Milestone 3	SkyAssist: Multimodal airline support agent	Gradio, GPT-4o, Whisper
21	Hugging Face Ecosystem	Model Hub, Inference API, Spaces deployment	HF Hub, Gradio, Spaces
22	Open-Source LLMs	Llama 4, Mistral, Phi-4, DeepSeek R1, Qwen 3	Ollama, vLLM, LM Studio
23	Audio AI & Whisper v3	Transcription, diarisation, voice agents	Whisper v3, pyanote
24	Code Generation AI	Python-to-C++ transpiler, profiling	GPT-5, cProfile, Hyperfine
25	Milestone 4	MeetScribe: Audio meeting intelligence	Whisper v3, Claude, Gradio
26	Evaluation & Tracing	LLM-as-judge, LangSmith, Phoenix dashboards	LangSmith, TruLens, Arize
27	Safety & Guardrails	Prompt injection, PII redaction, Llama Guard 3	NeMo, Presidio
28	FastAPI, Docker, CI/CD	REST + WebSocket, containerisation, GH A	FastAPI, Docker, GH A
29	Phase 1 Review Sprint	Code review, test coverage, mkdocs site	pytest, mkdocs

Day	Topic	Key Tasks	Tools
30	Milestone 5	CodeXcelerate: Python-to-C++ transpiler	GPT-5, Clang, OpenMP

★ Milestone 3 (Day 20): SkyAssist

Multimodal airline support agent: flight search, booking lookup, baggage Q&A, boarding pass scanning via GPT-4o vision, voice input via Whisper, Gradio UI.

Stack: GPT-4o, Gradio, Whisper, FastAPI.

★ Milestone 4 (Day 25): MeetScribe

Audio meeting intelligence. Whisper transcription, speaker diarisation, LLM summarisation comparing GPT-5 vs Llama 4 vs Claude Opus 4.7, PDF export, Slack notification.

Stack: Whisper v3, pyannote, Claude Opus 4.7, Llama 4, Gradio.

★ Milestone 5 (Day 30): CodeXcelerate

Python-to-C++ AI transpiler achieving up to 60,000x speedup. Multi-step LLM pipeline with OpenMP parallelism and Hyperfine benchmarks.

Stack: GPT-5, Clang, OpenMP, Hyperfine, Streamlit.

5 Phase 2: LLM Engineering (Days 31–60)

Phase 2 Focus Areas

Frontier Models • Fine-tuning (LoRA, QLoRA, DPO, ORPO) • Quantisation • Multimodal • Evaluation • Cloud Deployment

Day	Topic	Key Tasks	Tools
31	2026 LLM Landscape	GPT-5, Claude Opus 4.7, Gemini 2.5, Llama 4 benchmarks	LMSys Arena, HELM
32	Tokenisation Deep Dive	BPE, WordPiece, SentencePiece, context windows up to 2M	tiktoken, HF tokenizers
33	Advanced Prompting	Tree-of-Thought, self-consistency, prompt caching	DSPy, Outlines
34	Structured Outputs	JSON mode, grammar-constrained decoding, Pydantic AI	Outlines, Instructor
35	Milestone 6	PriceOracle: Predict prices with frontier LLMs	GPT-5, Claude, Gemini
36	Fine-tuning Foundations	LoRA, QLoRA, adapters, dataset preparation	PEFT, bitsandbytes
37	Supervised Fine-tuning	Fine-tune Llama 4 8B with Unsloth on domain data	TRL, Unsloth, Axolotl
38	Preference Tuning	DPO, ORPO, KTO, GRPO, reward modelling	TRL DPOTrainer, W&B
39	Quantisation & Inference	GPTQ, AWQ, GGUF, vLLM, SGLang serving at scale	llama.cpp, vLLM, SGLang
40	Milestone 7	DealHunter: QLoRA-tuned open-source predictor	QLoRA, Llama 4, TRL
41	RAG vs Fine-tuning	Decision framework, hybrid approaches	RAGAS, W&B Weave
42	Multimodal Models	Vision-Language: GPT-5, Claude Vision, Gemini Vision	GPT-5, LLaVA, CLIP
43	Embeddings at Scale	ANN at production scale, pgvector, Qdrant, Voyage 3	Qdrant, Weaviate
44	LLM Evaluation	MT-Bench, MMLU-Pro, GPQA, custom eval harness	HELM, lm-eval-harness
45	Milestone 8	BargainBot: Multi-agent deal-finder with alerts	CrewAI, Twilio, Celery
46	Observability	LangSmith tracing, Phoenix, OpenTelemetry	LangSmith, Arize Phoenix

Day	Topic	Key Tasks	Tools
47	Safety, Governance, Red Teaming	Llama Guard 3, EU AI Act, India DPDP	NeMo, Presidio, Garak
48	API Design & Gateway	SSE streaming, OpenAI-compatible API, LiteLLM proxy	FastAPI, LiteLLM
49	Cloud Deployment	AWS Bedrock, GCP Vertex AI, Azure AI Foundry	AWS, GCP, Azure, Modal
50	Milestone 9 (NEW)	VoiceForge: Real-time voice agent (sub-300ms)	ElevenLabs, Cartesia
51–60	Phase 2 Review & R&D	Refactor, document, research exploration	All Phase 2 tools

★ Milestones 6 & 7 (Days 35 & 40): PriceOracle + DealHunter

PriceOracle: predict product prices using GPT-5, Claude Opus 4.7, Gemini 2.5 Flash. Compare MAE/MAPE on live Streamlit leaderboard.

DealHunter: QLoRA fine-tune Llama 4 8B to compete with frontier models. Target sub-10% MAPE at less than \$0.001 per inference.

★ Milestone 8 (Day 45): BargainBot

CrewAI-powered system with 3 agents (Scanner, Analyst, Negotiator) monitoring e-commerce sites for price drops. SMS/Slack alerts via Twilio. Celery scheduling with PostgreSQL persistence.

★ Milestone 9 NEW (Day 50): VoiceForge Real-Time Voice Agent

Sub-300ms latency voice agent handling end-to-end conversations. Streaming STT (Whisper v3), LLM (GPT-5 or Claude Opus 4.7), TTS (ElevenLabs or Cartesia Sonic). Deploy as phone agent via Twilio Voice.

Stack: OpenAI Realtime API, ElevenLabs, Cartesia, Twilio, FastAPI WebSockets.

6 Phase 3: Agentic AI Systems (Days 61–90)

Phase 3 Focus Areas

ReAct • Reflexion • LangGraph 1.0 • CrewAI • AutoGen • MCP Servers • Browser & Computer Agents • Multi-Agent Systems

Day	Topic	Key Tasks	Tools
61	Agent Foundations	ReAct, Reflexion, Plan-and-Execute, tool use	LangChain Agents
62	MCP Protocol Deep Dive	Build custom MCP servers, security, auth	Anthropic MCP SDK
63	LangGraph 1.0	State machines, conditional edges, checkpoints	LangGraph 1.0
64	Browser Agents	Playwright agents, autonomous navigation	Playwright, Stagehand
65	Milestone 10	TalentTwin: Career Digital Twin	LangGraph, FastAPI
66	CrewAI Framework	Crew, Agent, Task, Process, role-playing	CrewAI
67	SDR Sales Agent	Email-writing with CRM, lead enrichment	CrewAI, SendGrid
68	Deep Research Agent	Planner + Searcher + Writer + Editor team	CrewAI, Tavily, Exa
69	Stock Picker Agent	SEC filings, news sentiment, signals	CrewAI, yfinance
70	Milestone 11	DevSquad: 4-agent engineering team	CrewAI, Docker
71	AutoGen v0.4	Conversable agents, group chat, nested	AutoGen v0.4
72	Agent Creator Agent	Meta-agent building new agents	AutoGen, Docker
73	Computer Use Agents	Claude Computer Use, Operator-style	Claude Computer Use
74	Multi-Agent Communications	A2A protocol, message queues, events	RabbitMQ, Redis Streams
75	Milestone 12	SidekickBrowser: Operator clone	LangGraph, Playwright
76	Advanced MCP	3 custom MCP servers: DB, REST, Filesystem	MCP SDK, FastAPI
77	Agent Memory Systems	Episodic, semantic, procedural memory	Mem0, Zep, Letta

Day	Topic	Key Tasks	Tools
78	Agent Evaluation	AgentBench, GAIA, SWE-bench Verified	AgentBench, Lang-Smith
79	Production Hardening	Rate limiting, circuit breakers, fallbacks	Tenacity, LiteLLM
80	Milestone 13	AgentForge: Meta-agent creating agents	AutoGen, Docker
81–83	Grand Capstone	AlphaTrader: 4-agent + 6 MCP + 44 tools	All Phase 3 tools
84–87	Docker, k8s, Helm	Multi-container, Helm charts, CI/CD	Docker, k8s, Helm
88–89	Docs & Final Tests	Test suite, Swagger docs, user manual	pytest, mkdocs
90	Final Presentations	20-min live demo to panel + Q&A	All projects

★ Milestone 10 (Day 65): TalentTwin

AI agent representing you to employers. Autonomously scrapes GitHub, LinkedIn, portfolio. Responds to recruiter queries. Public deployment with Calendly integration.

Stack: LangGraph, Playwright, FastAPI, Railway, Calendly API.

★ Milestone 11 (Day 70): DevSquad

Project Manager, Developer, Code Reviewer, QA Tester agents in isolated Docker containers. Fully autonomous from specification to tested code with GitHub integration.

Stack: CrewAI, Codex, Docker, GitHub API, pytest.

★ Milestone 12 (Day 75): SidekickBrowser

OpenAI Operator-class sidekick. Fills forms, navigates UIs, extracts structured data, executes multi-step web tasks from plain English. Chrome extension with FastAPI backend.

Stack: LangGraph, Playwright, FastAPI, Chrome Extension, SSE.

★ Milestone 13 (Day 80): AgentForge

Takes natural language description, generates agent code, tests it in sandboxed Docker, deploys as live running service fully autonomously.

Stack: AutoGen v0.4, Docker, FastAPI, GPT-5, GitHub Actions.

★ Grand Capstone (Days 81–90): AlphaTrader

Complete trading floor with 4 autonomous agents, 6 MCP servers, 44 tools.

Agents:

- **MarketAnalyst:** real-time prices, news sentiment, SEC filings
- **QuantStrategist:** statistical arbitrage, momentum, portfolio optimisation

- **RiskGuardian:** position limits, drawdown rules, compliance

- **ExecutionEngine:** order placement via broker API (paper trading)

6 MCP Servers: Market Data, News Feed, Financial Statements, Order Management, Risk Engine, Notifications.

Stack: AutoGen, MCP, LangGraph, Alpaca API, Grafana, Docker, Kubernetes, FastAPI.

7 Post-90 Placement Bridge

The 2026 edition includes a structured **Placement Bridge** after Day 90 to convert your training into a real job offer. The bridge spans 4–6 weeks and is mandatory for the *Excellence* grade.

7.1 Bridge Activities

➡ Week 1: Portfolio Polish & Public Presence

- Pin top 5 projects on GitHub with professional READMEs and demo videos
- Deploy 2 flagship projects publicly (Hugging Face Spaces, Vercel, Modal)
- Publish 2 technical blog posts on Medium or Hashnode
- Update LinkedIn with all projects, request cohort endorsements
- Submit 1 talk proposal to a meetup or conference

➡ Week 2: Resume & Cold Outreach

- ATS-optimised resume in three variants (ML, LLM/Agentic, Data Engineering)
- Tracker of 50 target companies and 100 named contacts
- Send 25 personalised cold emails per week to hiring managers
- Apply to 30 roles on LinkedIn, Wellfound, YC Work
- Engage on LinkedIn and X with 3 substantive posts per week

➡ Week 3: Interview Preparation

- DSA refresher: 50 LeetCode mediums on arrays, graphs, DP
- ML systems design: feature stores, retraining loops, drift detection
- LLM systems design: RAG, agentic workflows, cost vs latency
- Behavioural prep: STAR stories for every milestone project
- Mock interviews: 3 technical + 2 behavioural with mentors

➡ Weeks 4–6: Take-Homes, Onsites, Negotiation

- Take-home templates: RAG eval harness, fine-tuning report, agent demo
- Onsite simulation: coding, system design, ML/LLM design, behavioural
- Salary benchmarking via levels.fyi, AmbitionBox, Glassdoor
- Offer negotiation: counter offers, equity, sign-on bonus
- Reference letters from AlgoProfessor leadership for top performers

7.2 Bridge Milestones

★ Bridge Milestone A: Open-Source Portfolio PR

Contribute a meaningful pull request to one of: LangChain, LangGraph, CrewAI, AutoGen, vLLM, Unsloth, or any high-star agentic AI repository. The PR must be merged or have substantive maintainer engagement.

★ Bridge Milestone B: Final Showcase Demo Day

Live 15-minute demo to a panel of hiring managers from AlgoProfessor's industry network. Panellists may extend interview invitations on the spot. Top 10% receive a paid AlgoProfessor Research Associate offer.

8 Jobs After Completion: 2026 Market Intelligence

The AI talent market in 2026 is the strongest it has ever been. The 18 milestones map directly to roles companies are actively hiring for. Salary figures are indicative based on publicly reported ranges.

8.1 Primary Roles & Salary Bands (2026)

Role	Skills Match	India (LPA)	USA (USD)
AI / ML Engineer	Phase 1 + Milestones 1, 5, 6	12–35 L	140–220 K
LLM Application Engineer	Phase 1 & 2 + Milestones 3, 4, 7, 8	18–45 L	160–260 K
Agentic AI Engineer	Phase 3 + Milestones 10–13, Capstone	22–55 L	180–320 K
RAG / Search Engineer	Milestones 2, 3, 8 + Pinecone, Neo4j	15–40 L	150–240 K
Fine-tuning Specialist	Phase 2 + Milestones 7, 8 + W&B	18–50 L	170–280 K
MLOps / LLMOps Engineer	Days 26, 28, 46, 84–87 + Capstone	16–42 L	160–250 K
AI Safety / Red Team Engineer	Days 27, 47 + Garak + Llama Guard 3	18–48 L	180–300 K
Voice / Multimodal Engineer	Days 23, 42, Milestone 9	20–50 L	180–290 K
AI Research Associate	Capstone + paper + open-source PRs	12–30 L	130–210 K
Data & Analytics Engineer	Days 3, 11–12 + Pandas, Polars, SQL	10–28 L	120–180 K

8.2 Companies Actively Hiring in 2026

➤ Indian Product Companies & Startups

AI-first: Sarvam AI, Krutrim, Soket AI Labs, Ola AI Research, Adya, Sprinklr AI, Glance AI, EkStep, AI4Bharat.

Unicorns: Flipkart, Swiggy, Zomato, Razorpay, CRED, PhonePe, Postman, Freshworks, Zoho, Paytm.

Healthcare AI: Niramai, Qure.ai, SigTuple, HealthifyMe, Practo AI, Tata 1mg.

FinTech AI: Zerodha, Groww, Upstox, Smallcase, AngelOne, INDmoney.

➔ Global Tech & AI Companies with India Centres

Frontier labs: Anthropic, OpenAI, Google DeepMind, Meta AI, Mistral AI, Cohere, xAI.

Big tech India: Google, Microsoft, Amazon, Apple, Meta, Nvidia, IBM, Adobe, Salesforce.

Enterprise AI: Databricks, Snowflake, Confluent, MongoDB, Elastic, Datadog.

AI infra: Together AI, Modal, Replicate, Anyscale, Lambda, RunPod, Hugging Face.

➔ IT Services & GCC AI Practices

Indian IT majors: TCS Research, Infosys Topaz, Wipro AI, HCL AI Labs, Tech Mahindra Makers Lab, LTIMindtree Canvas.AI, Mphasis AI.

GCCs: JPMorgan, Goldman Sachs, Morgan Stanley, Wells Fargo, American Express, Walmart Labs, Target India.

Consulting: Deloitte AI Institute, Accenture AI, EY AI, KPMG Lighthouse, PwC AI.

➔ Research Labs & Academic Roles

Indian labs: TIFR, IISc, IIT AI labs, IIIT-H, ISI Kolkata, Wadhvani AI, AI4Bharat.

Global: MILA Montreal, Vector Institute, ETH AI, EPFL, MIT CSAIL, Stanford HAI.

AlgoProfessor partners: BITS Pilani (MoU), IJCMi editorial board, university research collaborations.

8.3 Three Career Paths

Path 1: Industry Engineer Track

Target roles: AI/LLM/Agentic AI Engineer at product companies or startups.

Focus: Phase 3 milestones, system design, production deployment.

Timeline to offer: 4–8 weeks post-completion.

Compensation: INR 18–45 LPA in India, USD 160–280K in USA.

Path 2: Research & Academia Track

Target roles: Research Engineer, PhD admission, postdoctoral position.

Focus: 1–2 conference papers, Capstone with novel contribution, recommendation letters.

Timeline: 6–12 months for PhD admission, 3–6 months for research engineer.

Compensation: INR 8–25 LPA, USD 110–210K.

Path 3: Entrepreneurship Track

Target outcome: Found an AI startup or join early-stage YC company.

Focus: TalentTwin, AgentForge, AlphaTrader as portfolio anchors; investor intros.

Outcome: pre-seed funding INR 50L–3Cr; AlgoProfessor takes no equity.

9 Evaluation & Grading

Assessment Component	Points	Evaluated By
Daily GitHub submissions (timeliness + quality)	10	Automated + Mentor
Code quality, PEP 8, documentation, tests	12	Mentor Code Review
Phase 1 Milestones (5 projects)	20	Technical Panel
Phase 2 Milestones (5 projects)	18	Senior Engineers
Phase 3 Milestones (4 projects)	15	AlgoProfessor Team
Grand Capstone: AlphaTrader	10	CEO + Board Panel
Placement Bridge milestones	10	Industry Panel
Peer collaboration & professionalism	5	Team + Mentor
Total	100	

Score	Grade	Recognition
90–100	Excellent	Certificate + LoR + LinkedIn Endorsement + Research Associate offer (top 10%)
75–89	Good	Completion Certificate + LinkedIn Endorsement + Placement Bridge
60–74	Satisfactory	Participation Certificate + Mentor debrief
Below 60	Incomplete	Review with supervisor; programme may be terminated

10 Complete Technology Stack 2026

The programme covers 60+ tools and frameworks across 16 categories spanning the entire modern AI engineering stack, from version control to production-grade Agentic AI deployment.

Version Control & DevOps

Git, GitHub, GitHub Actions, Docker, Kubernetes, Helm, pre-commit, mkdocs, pytest, uv, ruff.

Deep Learning

PyTorch 2.x, torch.compile, FSDP, torchvision, timm, TensorBoard, ONNX, Albumentations.

Python Scientific Stack

Python 3.12+, NumPy, Pandas 2.x, Polars, SciPy, Matplotlib, Seaborn, Plotly, Dash, DuckDB, PyArrow.

Databases & Storage

PostgreSQL, SQLAlchemy 2.0, MongoDB Atlas, motor, Redis, SQLite, pgvector, MinIO, DuckDB.

Machine Learning

Scikit-learn, XGBoost 2.x, LightGBM, CatBoost, SHAP, Optuna 4.x, UMAP, Featuretools.

Vector Stores

Chroma, FAISS, Pinecone, Qdrant, Weaviate, Milvus, Voyage AI, Cohere Embed v4.

LLM APIs (2026)

OpenAI GPT-5, Anthropic Claude Opus 4.7, Google Gemini 2.5, xAI Grok 3, Mistral, Cohere, LiteLLM.

Voice & Multimodal

Whisper v3, ElevenLabs, Cartesia Sonic, OpenAI Realtime, pyannote, LLaVA 1.6, CLIP, SAM 2, DI-NOv3.

Open-Source LLMs

Llama 4, Mistral, Phi-4, DeepSeek R1, Qwen 3, Gemma 3, vLLM, SGLang, Ollama, llama.cpp.

Web, API & UI

FastAPI, Streamlit, Gradio, SSE, WebSockets, Swagger, Chrome Extensions, Next.js, Vercel.

Fine-tuning

LoRA, QLoRA, PEFT, TRL (SFT, DPO, ORPO, KTO, GRPO), bitsandbytes, Unsloth, Axolotl, W&B.

Observability & Eval

LangSmith, Arize Phoenix, TruLens, Weave, OpenTelemetry, MLflow, lm-eval-harness, Agent-Bench, Garak.

RAG Frameworks

LangChain, LlamaIndex, Haystack, RAGAS, Cohere Rerank, Unstructured.io, LlamaParse, GraphRAG.

Safety & Governance

NeMo Guardrails, Llama Guard 3, Presidio PII, Garak red-teaming, EU AI Act, India DPDP.

Agentic Frameworks

LangGraph 1.0, CrewAI, AutoGen v0.4, Pydantic AI, smolagents, MCP SDK, Mem0, Zep, Letta.

Cloud & Integrations

AWS Bedrock, GCP Vertex AI, Azure AI Foundry, Modal, Replicate, Together AI, Twilio, SendGrid, Slack.

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